

Lab: Thought Experiment — Communication, Other Minds, and Shared Meaning

Objective

Investigate the philosophical foundations of communication through thought experiments on intersubjectivity, shared meaning, and the limits of understanding.

Part 1: The Translation Problem

Goal: Explore the limits of communication across divergent generative models.

Imagine two speakers who share no common language and no common cultural background. They must communicate a simple message: "There is danger ahead."

- What minimal resources must they share for communication to succeed? (Shared world? Shared embodiment? Shared attention?)
- How does Active Inference frame this problem? (Hint: coupled inference requires some initial overlap in generative models.)
- Does the requirement for shared generative structure mean that fully universal communication is impossible? Or is embodied interaction sufficient?

Write your response here

Part 2: Theory of Mind in Action

Goal: Analyze the recursive structure of social cognition.

Consider this scenario: Alice believes that Bob believes that she is lying. Actually, she is telling the truth.

- Map out the recursive hierarchy of beliefs. How many levels of Theory of Mind are required?
- How does Active Inference model this recursive embedding (agent A's generative model contains a model of agent B's model of agent A)?
- At what point does the recursion become computationally intractable? Does this explain why real-world communication sometimes fails spectacularly?

Write your response here

Part 3: Metaphor and Creative Communication

Goal: Analyze non-literal communication through the lens of Active Inference.

When someone says "time is money," they are not stating a literal fact but creating a mapping between two domains.

- How does metaphor work from an Active Inference perspective? (Hint: the listener's generative model is extended by applying structure from one domain to another.)
- Is metaphor a failure of precise communication or its highest achievement?
- Construct a novel metaphor for the concept of communication itself, drawing on Active Inference vocabulary.

Write your response here

Part 4: The Ethics of Communication

Goal: Examine the ethical dimensions of coupled inference.

Active Inference shows that communication can be used to synchronize generative models (genuine communication) or to manipulate them (deception, propaganda).

- What makes deception possible in the Active Inference framework? (Hint: the deceiver must model the target's model to predict how the false signal will be interpreted.)
- Is there a principled distinction between persuasion and manipulation? Can Active Inference formalize it?
- How does this connect to Habermas's concept of "communicative rationality" — the ideal of communication oriented toward mutual understanding rather than strategic advantage?

Write your response here

Part 5: Synthesis and Position Defense

Goal: Articulate your philosophical view on communication.

Write a 200-word statement on one of:

- "Communication is fundamentally cooperative — the Active Inference framework shows that mutual understanding is the default, and deception is parasitic on it."
- "Language is the most powerful niche construction technology ever invented — it restructures not just the environment but other agents' generative models."
- "The problem of other minds is dissolved, not solved, by Active Inference: we never access other minds directly, but we don't need to."

Share with a peer and defend against their strongest objection.

Write your response here

Lab Summary

Part	Skill Developed	Key Philosophical Move
1	Limit analysis	Identifying the minimal conditions for cross-model communication
2	Recursive reasoning	Mapping the hierarchical structure of Theory of Mind
3	Creative analysis	Analyzing metaphor as generative model extension
4	Ethical reasoning	Distinguishing persuasion from manipulation under Active Inference
5	Position defense	Constructing and defending a philosophical position on communication